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Analytic Deep Learning and Stepwise Integrated Gradients-based Power System Transient Stability Preventive Control

Youbo Liu, *Member, IEEE*, Tong Su, *Student Member, IEEE*, Gao Qiu, Hongjun Gao, *Senior Member, IEEE*, Junyong Liu, *Member, IEEE*, Yue Shui

***Abstract*—Trajectory sensitivity (TS) is a powerful alternative to linearize strong-nonlinear transient stability patterns. However, due to its incomplete view of an intact power system operational space, it can be readily to drive preventive control to stuck in local stationary solution. To mitigate this issue, an integrated gradient (IG)-based stepwise control method is proposed. The key idea is to develop IG to enable a control sensitivity taking both merits of local and global sensitivity, further help preventive control escape from local stationary points. Particularly, deep belief network (DBN) is firstly utilized to parameterize pattern between transient stability index (TSI) and steady-state dispatches. It is then leveraged to infer IG to linearize transient stability constraints in real-time. Interior point method is employed to solve the linearized model. A stepwise control strategy is finally proposed to adaptively tune the integrated interval of the applied IG, such that the preventive control path can be ensured to follow the trend of improving transient stability. Case studies on two benchmarks show that the proposed method outperforms traditional TS method in terms of efficiency and economy, and provides a promising way to enhance engineering sensitivity methods.**

***Index Terms*—Deep learning, integrated gradient, transient stability preventive control, surrogate model.**

NOMENCLATURE

A. Sets

$\mathbf{S_T}$	The set of synchronous machines.
$\mathbf{S_W}$	The set of wind farms.
$\mathbf{S_D}$	The set of loads permitted to be shed.
$\mathbf{S_B}$	The set of buses.
$\mathbf{S_L}$	The set of transmission lines.
$\mathbf{S_S}$	The set of variables that are involved in sensitivity computation. In usual, $\mathbf{S_S} = \mathbf{S_T} \cup \mathbf{S_W}$. If load shedding is allowed, $\mathbf{S_S} = \mathbf{S_T} \cup \mathbf{S_W} \cup \mathbf{S_D}$.

B. Parameters/ Indices

m	The quantities of the interpolation points within the integrated interval for integrated gradient computation.
N_c	The total number of controllable resources.
N_T	The number of the samples in the testset.
G_{ij}/B_{ij}	Conductance and susceptance located in *i*th row and *j*th column of admittance matrix $\mathbf{Y}$.
$\Delta P_g^{\min}/\ \Delta P_g^{\max}$	The down-ramp limit of g-th generator / The up-ramp limit of g-th generator.
$P_{Gi}^{\min}/\ P_{Gi}^{\max}$	The lower/upper bound of active generation of g-th generator.
$Q_{Gi}^{\min}/\ Q_{Gi}^{\max}$	The lower/upper bound of reactive generation of g-th generator.
$V_i^{\min}/\ V_i^{\max}$	The lower/upper bound of voltage at bus i
$P_l^{\min}/\ P_l^{\max}$	The lower/upper bound of active power flow through branch l.
$P_d^{\min}/\ P_d^{\max}$	The lower/upper bound of load shedding of the d-th load permitted to be shed.
TSI_C	The pre-set threshold of transient stability margin.
TSI_Δ	True critical TSI identifying stability. If the tolerable angular difference maximum is 360°, $\mathrm{TSI}_\Delta = 0$. $\mathrm{TSI} > \mathrm{TSI}_\Delta$ infers transient stability.
Δ	The pre-set convergence threshold for the proposed stepwise control scheme.

C. Variables

$\mathbf{\Lambda}$	The input features of DBN.
TSI_P	The output of DBN, i.e., the worst TSI under multi-contingency.
TSI^0	The worst-contingency TSI computed at an initial steady-state equilibrium operating condition.
S_{Si}	The ith local sensitivity of stability margin against the ith controllable variable.
IG_i	The IG-based sensitivity of stability margin against the ith controllable variable.
$\mathbf{\Lambda_C}$	The controllable variables/resources.
$\Lambda_{\mathrm{C},i}$	The ith controllable variable in $\mathbf{\Lambda_C}$.
$\mathbf{\Lambda}^0, \mathbf{\Lambda}_\mathrm{C}^0$	The input features collected from a steady-state equilibrium operating condition, and its controllable sub-feature set.
$\epsilon^k, \mathbf{IG}^k$	The bound of integrated interval at the kth control loop, and the integrated gradient vector computed based on this interval.
$\mathbf{\Lambda}_\mathrm{C}^{\mathbf{P},k}$	The control actions made by IPM solving the TSCOPF whose constraint (12) is linearized by $\mathbf{IG}^k$.

D. Functions

$\mathcal{R}(\cdot)$	The DBN-based surrogate. A nonlinear mapping with neural network structure.

$\mathbb{E}[\cdot]$	The function to compute expectation.
$C_G(\cdot)/C_W(\cdot)$	The control cost functions of synchronous machines/wind farms.
$C_L(\cdot)$	The cost function of load shedding. This function works only when load shedding is permitted.

E. Abbreviation

DBN	Deep belief network.
IG	Integrated gradient.
TS	Trajectory sensitivity.
TSI	Transient stability index.
TDS	Time-domain simulation.
TSA	Transient stability assessment.
DL	Deep learning.
TSCOPF	Transient stability constrained optimal power flow.
IPM	Interior point method.
3ϕ	Three-phase.

I. Introduction

With higher penetration of intermittent and stochastic renewable energy, lager scale of grid interconnection and less power reserves, power system operation is closer to its stability limits than ever. Estimation and preventive control tools for transient stability are thus in urgent demand of efficiency and accuracy [1].

Time-domain simulation (TDS) is fundamental to transient stability assessment (TSA). It enables accurate modelling for dynamics but is computationally intensive [2]. Towards faster TSA, some energy-based methods, e.g., extended equal-area criterion [3], Kalman filtering techniques [4], Lyapunov method [5], sparsity techniques [6], nonparametric methodology [7], etc., are proposed. These methods are however highly dependent on accurate post-fault trajectory measurements and can be in trouble with multi-contingency pre-fault analysis. To harness high-dimensional pre-fault TSA, many data-driven methods are burgeoning, e.g., support vector machine [8], principal component analysis [9], deep belief network (DBN) [10], autoencoder [11] and convolutional neural network [12]. Among them, deep learning (DL) family significantly outperforms others with regard to (w.r.t.) generalization, thus is the best alternative for fitting highly nonlinear TS patterns.

When instability risk is identified, preventive control must be engaged. The concept of preventive control refers to risk-averse actions for latent contingencies [13], and it usually formed as transient stability constrained optimal power flow (TSCOPF). But TSCOPF features high-dimensionality and nonlinearity due to inclusion of large sets of differential-algebraic equations (DAEs), thus is time-consuming to be solved by traditional solvers [14]. Improvements are then motivated, such as trajectory sensitivity (TS) method [15], surrogate modelling [10], [16], hybrid decision trees [17], deep reinforcement learning [18] and data-driven sensitivity method [19], [20]. [17] and [18] both only used pure data-driven models, and are of poor generalization. A better idea is to incorporate data and physic models, as [10] and [16] proposed. However, as surrogate modelling is still nonlinear and nonconvex system, convergence of such technique is hard to be theoretically proved. Considering this, research using linearization to simplify transient stability model are carried out. By means of TS, [15] locally linearize control sensitivity from post-fault trajectories at a specified steady-state equilibrium point. Despite it improves solving efficiency, accuracy is sacrificed. Exactly as in [21], local sensitivity is primarily limited to first or second-order approximations around operational or equilibrium conditions, and it cannot properly account for the impact of the full variation range of the uncertainty sources (e.g., large changes of power generation and consumption), as well as model nonlinearities. If the exploration of entire stochastic space is necessary, and model linearity assumptions are too strong or unrealistic to hold, local sensitivity reveals to encounter with serious distortion of true operational patterns of power systems, likely turning into a quite perfunctory and imprecise activity. [19] and [20] used data-driven sensitivity methods to promote generalization of TS against shifting operations, but their local sensitivity can still loss true evolution of transient stability. This issue comes from many causes, such as gradient saturation induced by learners, local stationary point, etc. An intuitive explanation is that, in a stationary point, local sensitivity of a controllable variable can be 0, whose true tendency to vary transient stability is nontrivial.

In light of the above considerations, this paper proposed an analytic deep learning and stepwise integrated gradients-based transient stability preventive control. Its merits are four folds:

1) Analytic deep learning is utilized to estimate local sensitivity of transient stability versus control variables. It allows us to achieve real-time sensitivity analysis, and surrogate transient stability constraints. In this way, time-consuming trajectory analysis under multi-contingency can be bypassed.

2) For the first time, integrated gradient (IG) technique is conducted to enhance local sensitivity. This method is partially benefited by global sensitivity, it thus, on one hand, fixes the issue of overlooking true security dynamics by local sensitivity (see [15], [19], [20]), on the other hand, circumvents erroneous estimation for control sensitivities due to gradient saturation of learning models [22]. This key idea supports us to fast and accurately seek out optimal control path.

3) A stepwise control scheme is proposed to enable reliable and effective IG-guided preventive action. Although global sensitivity idea synthesized in IG method is immune to stationary points, it is unable to reveal control quantities at a dispatch. We thus designed a stepwise dichotomy to adaptively tune integrated interval of IG, then combined interior point method to decide control actions in a close loop manner.

4) What is striking is that IG is verified to not only be able to interpret deep learning, but also to enable preventive control for complex system stability. By trial, IG outperforms traditional

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TS methods on both control cost and efficiency.

The rest of the paper is organized as follows. Section II presents integrated gradient method. Section III renders the proposed preventive control scheme. Case studies are given in Section IV. Section V concludes the paper.

II. Integrated Gradients-based Sensitivity of Transient Stability Against Generators or Loads

Before describing the proposed method, we first provide an important definition on control sensitivity, which will be helpful for understanding our work.

Definition 1 (one-point sensitivity or local sensitivity): we define control sensitivity that is computed at a specified steady-state operational equilibrium point as one-point sensitivity or local sensitivity.

Definition 2 (IG-based sensitivity): we define control sensitivity that is expectation of one-point sensitivities over the vicinity of a specified steady-state operational equilibrium point as IG-based sensitivity, as this paper proposes integrated gradient (IG) technique to calculate this sensitivity.

Evidently, IG-based sensitivity equals to the expectation of one-point sensitivities under the above definitions. For a specified operational point $\boldsymbol{x}$, IG-based sensitivity can be roughly represented as $IG_{\boldsymbol{x}} = \mathbb{E}_{\Delta\in B_\epsilon}[\nabla_{\boldsymbol{x}+\Delta}]$, where $\nabla_{\boldsymbol{x}+\Delta}$ stands for one-point sensitivity at $\boldsymbol{x}+\Delta$ point, B_ϵ is a hyper-polyhedron vicinity enveloped by small perturbation ϵ on $\boldsymbol{x}$, $\Delta\in B_\epsilon$ denotes that perturbations on $\boldsymbol{x}$ are within B_ϵ, and $\mathbb{E}_{\Delta\in B_\epsilon}$ is expectation operator. We will show details in the following sections.

Additionally, local sensitivity used in TSCOPF is computed by TS method, which has been widely investigated in preceding research. To highlight our method, we refer the readers to [15], [23] for details of TS, though a noteworthy point is that TS is intensively time-consuming to compute as it demands repeated TDS when multi-contingency is necessitated.

A. Surrogate-based Sensitivity Calculation

As physics-based sensitivity calculation of transient stability versus control variables is tremendously computational burdensome with multi-contingency involved, a wise alternative is surrogate-based way [24]. Surrogate intends to parameterize in-depth security patterns from massive transient responses. It technically models a direct mapping between transient stability margin and steady equilibrium point (in other words, power system dispatch). In this paper, DL is adopted as the surrogate in terms of accuracy and robustness.

Since our dataset appertains to 2-dimensional (2D) array, fully connected DL is preferred to process such type of data. DBN thus becomes our preference, as it has been widely justified to outperform others [10]. For details of DBN learning, we refer readers to [25]. But one thing should be noted is data preparation, as it is a decisive factor for DBN generalizability.

To prevent loss of generalization to system dynamics, training data must be carefully prepared to cover possible operations. To wholly represent system state, input features are determined as $\boldsymbol{\Lambda} = [\mathbf{V_T}, \mathbf{V_W}, \mathbf{P_T}, \mathbf{P_W}, \mathbf{P_L}, \mathbf{Q_L}]^{\mathrm{T}}$, encompassing voltage magnitudes of synchronous generators $\mathbf{V_G}$ and wind generators $\mathbf{V_W}$, active power of synchronous generators $\mathbf{P_T}$, wind generators $\mathbf{P_W}$ and loads $\mathbf{P_L}$, and reactive power of the loads $\mathbf{Q_L}$. Later for short, we denote $\mathbf{P_G} = [\mathbf{P_T}, \mathbf{P_W}]$. Target is transient stability index (TSI), which is an engineering index for evaluating transient stability and can be computed via (1).

$$\mathrm{TSI} = \frac{360^\circ - |\delta^{\max}|}{360^\circ + |\delta^{\max}|} \times 100 \tag{1a}$$

$$\delta^{\max} = \max\left(\delta_i - \delta_j\right), \forall i,j \in \mathbf{S_T} \tag{1b}$$

where δ_i is the rotor angle of ith generator; $\delta^{\max}$ is the maximum rotor angle difference between any two synchronous machines; $\mathbf{S_T}$ is the set of synchronous machines. TSI positively relates to transient stability level. [12] indicated an instability boundary of TSI = 0 (i.e., $\delta^{\max} = 360°$), whilst [26] provided a more conservative boundary of TSI = 33.3 ($\delta^{\max} = 180°$). In this paper, we adopt TSI^0 to denote initial transient stability margin under the worst contingency, and TSI_C to denote the set transient stability threshold. Notably, the latter is used to steer preventive control to avert control error and maintain conservativeness. We also introduce TSI_Δ to represent true critical TSI identifying transient stability, note that $\mathrm{TSI}_C \geq \mathrm{TSI}_\Delta$. TDS is used to acquire rotor angle trajectories so that to calculate TSI. Permanent 3-phase short-circuit fault on each line is selected as contingency. The label for DL is decided by worst-case TSI throughout contingencies. On the above feature definitions, Latin hypercube sampling (LHS) is introduced to produce training data, thanks to its ability to uniformly span over a whole prior space [10]. Detailly, LHS is performed to generate N_S steady-state equilibrium points. These points are then fed to TSI computation to label samples. Notably, to ensure conservativeness of the DL training, we introduce the trick of numerical perturbations to train the DBN that is further used to compute IG-based sensitivity. The numerical perturbation trick follows the idea that, DL is more trustworthy on the cases with high frequency of occurrence. So, a chi-square distribution is taken as the numerical perturbator, which produces larger negative offsets to those TSIs more likely to occur. This trick decreases TSIs for all samples and can tune the distribution of labels towards conservativeness. The process is clarified in Fig. 1. After sample collection, N_S samples are ready for DL training. On the other hand, cost sensitive learning proposed in [27] to train the follow-up DBN ensemble that is adopted for TSA. For conservative TSA, cost sensitive matrix is configured to prompt the DBN ensemble to tend to under-estimate stability margin. We skipped the details of DL training, as it is an old topic.

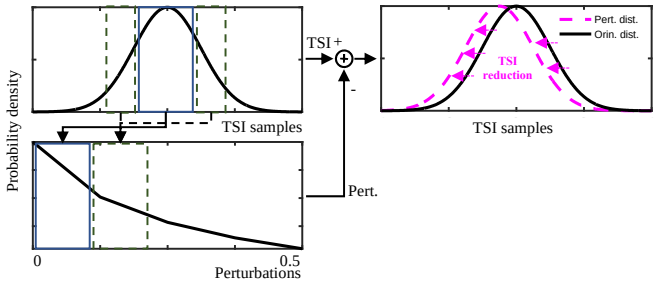


Fig. 1 Perturbation on raw TSI data for building conservative transient stability surrogate

After training, the DBN-based surrogate is then expressed as:

$$\mathrm{TSI}_P = \mathcal{R}(\boldsymbol{\Lambda}) \tag{2}$$

where $\mathcal{R}$ denotes the trained DBN; subscript *P* indicates prediction of TSI. (2) is high parallelable, which can be impressively accelerated by GPU and enables 1 second level TSA. Based on the prediction, transient stability can be inferred by (3):

$$\mathcal{R}(\boldsymbol{\Lambda}) > 0 \tag{3}$$

(2) also provide important sensitivity information of TSI against generators or loads, which can be obtained by calculating partial derivative of output of the DBN versus inputs:

$$S_{Si} = \frac{\partial \mathrm{TSI}_P}{\partial \Lambda_{\mathrm{C},i}} = \frac{\partial \mathcal{R}(\boldsymbol{\Lambda})}{\partial \Lambda_{\mathrm{C},i}}, i \in \mathbf{S_S}, \Lambda_{\mathrm{C},i} \in \boldsymbol{\Lambda_C} \subset \boldsymbol{\Lambda} \tag{4}$$

where $\boldsymbol{\Lambda_C}$ denotes the controllable features. In this paper, $\mathbf{P_G}$ and $\mathbf{P_L}$ are controlled by generator redispatch and load shedding, respectively, so $\boldsymbol{\Lambda_C} = [\mathbf{P_T}, \mathbf{P_W}, \mathbf{P_L}]^{\mathrm{T}} \in \boldsymbol{\Lambda}$. $\Lambda_{\mathrm{C},i}$ is the *i*th feature/variable in $\boldsymbol{\Lambda_C}$. S_{Si} is the sensitivity of the *i*th variable and $\mathbf{S_S}$ is the sensitivity collection of each control variable. We refer to [16] for detailed calculation process of (4).

Remark: (3) provides fast and accurate transient stability inference and can be handled through heuristic algorithms or nonlinear programming [16], [28]. However, heuristic methods are inefficient for large-scale problems, and nonlinear programming can be hard to convergence due to nonconvexity of (3). Using sensitivity to linear (3) can be a good idea, but it goes against the goal of control precision. Several research show that the control accuracy of sensitivity methods can be improved by introducing close-loop stepwise control scheme, where the whole solving path is divided into multiple successive iterations and sensitivity is updated at every iteration. This scheme asks for repeated TS calculations, thus is inefficient under multi-contingency constraints. Fortunately, this is not the case for DBN-based sensitivity computation. Following the above cogitation, the key challenge is gradient saturation of DBN, which is tackled by integrated gradient (IG). Interestingly, we found that IG can provide better sensitivity than TS method, which was analyzed in section II. B and testified in numerical study.

B. Integrated Gradients Calculation

IG is originally designed to evaluate importance of features. Differing from sensitivity calculation on local point, IG instead interprets contribution of a feature by integrating partial derivate over interval, such that gradient saturation can be well averted [22]. A further merit is that IG can more globally aware trend of transient stability varying with control actions. IG is thus introduced to enhance control sensitivity w.r.t. control variables (i.e., generation and permitted load shedding).

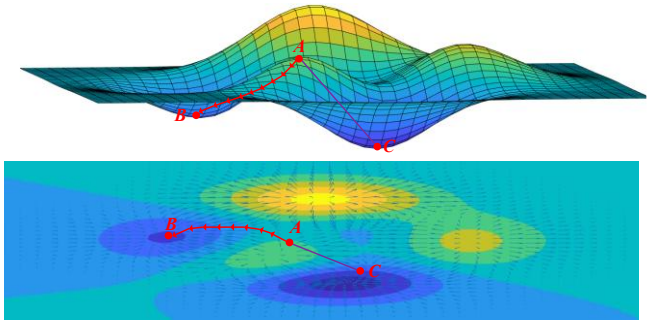
For brevity, we use superscript of 0 to denote initial unstable operating condition before control. Now suppose that $\boldsymbol{\Lambda_C}$ corresponds to control variables after control. For a specified variable in $\boldsymbol{\Lambda}_{\mathrm{c}}^{\mathbf{0}}$ (assume to be the *i*th one), we perturb it with a tiny positive value ϵ_i such that a neighborhood $\boldsymbol{\Lambda}_i^{\mathrm{P}}$ of the original input $\boldsymbol{\Lambda}^0$ can be attained. By means of (4), compute gradient integrated within the perturbation interval, IG of the *i*th control variable can be got. The mathematical formula of IG is given in (5):

$$\mathrm{IG}_i = \mathbb{E}\left[\int_{-1}^{1} \frac{\partial \mathcal{R}(\boldsymbol{\Lambda}^0 + \alpha \times (\boldsymbol{\Lambda}_i^{\mathrm{P}} - \boldsymbol{\Lambda}^0))}{\partial \Lambda_{\mathrm{C},i}} d\alpha\right], \forall i \in \mathbf{S_S} \tag{5}$$

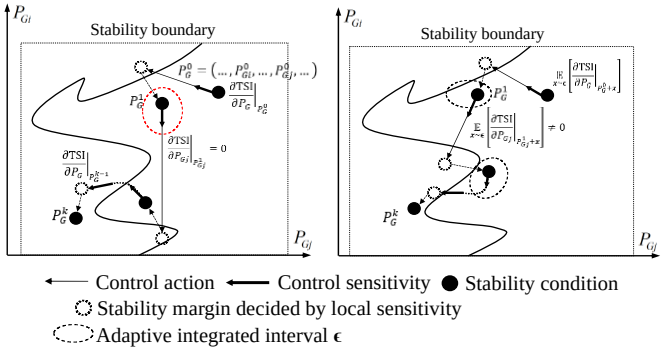
where $\boldsymbol{\Lambda}_{\mathrm{c}}^{\mathbf{0}}$ denotes the controllable features in $\boldsymbol{\Lambda}^0$, $\boldsymbol{\Lambda}_{\mathrm{c}}^{\mathbf{0}} \subset \boldsymbol{\Lambda}^0$; N_{c} is the size of $\boldsymbol{\Lambda_C}$, in other words, the number of controllable variables; $\mathbb{E}$ denotes the function of calculating mathematic expectation. In practice, (5) can be approximated through linear interpolation on the integrated path. This approximation finally results in the following equation:

$$\mathrm{IG}_i = \mathbb{E}\left[\sum_{k=-m}^{m} \frac{\partial \mathcal{R}(\boldsymbol{\Lambda}' + \frac{k}{m} \times (\boldsymbol{\Lambda}_i^{\mathrm{P}} - \boldsymbol{\Lambda}^0))}{\partial \Lambda_{\mathrm{C},i}}\right], \forall i \in \mathbf{S_S} \tag{6}$$

where *m* is the number of interpolation points, which is recommended in range from 20 to 300 [22].



(a) A 3D diagram to qualitatively visualize drawback of conventional local sensitivity method.



(b) Illustration of traditional local sensitivity control perspective (left subplot) and the proposed stepwise control strategy (right subplot). Note that, we here follow the implementation of conventional TS method, which uses local sensitivity to fully linearize stability constraints. This can incur error on transient stability assessment. In Fig. 2(b), we apply dashed circle to denote the stability margin decided by sensitivity, and solid circle to stand for the true stability margin at an iterative operation.

Fig. 2. Visualization of IG enhancing transient stability preventive control.

Compared with to common TS, the advantages of integrated gradients can be shown through Fig. 2. (2D analysis can be found in Fig. 4 and Fig. 5). The top plot in Fig. 2(a) is a general nonlinear 3-dimension plane, and the bottom one is the corresponding contour map. As transient stability boundary features alike pattern with such plane, we can depict the concept of integrated gradients by the lights of Fig. 2(a). Let's locate the hill where point A is placed. Assume that A is the stationary point, slope will be 0. This means, under traditional local TS workflow, controllable variables stop contributing to preventive actions. However, this is not the case for left-hand limit point of A, which still tends to take actions to go for optimum. Similar situation can be found in stationary point of a smooth asymmetric curve. In power system point of view, generators operating at such points could still promote system security. Although the contributions may be petty, they need to be dispatched in economical regards. It could be argued that, by integrating slopes or local sensitivities around a steady-state operation, we can

more accurately know true tendency of transient stability evolution, so that to benefit a more accurate and economical preventive control. This process will be more time-consuming compared to one-point sensitivity computation, fortunately, it can be significantly accelerated by DL-based IG.

Differences between traditional local sensitivity and IG-based sensitivity control can be evidently found in Fig. 2(b). In the left subplot of Fig. 2(b), P_{Gj} meets a local zero sensitivity at the control iteration corresponding to P_G^1. This incurs a attonity for the *j*th generator, which may do affect transient stability and be cost effective. Whilist the right subplot of Fig. 2(b) conveys that, IG-based method attempts to figure out what trend the sensitivity will evolve. In this case, it finds that P_{Gj} features trend of negative impact on transient stability, and therewith decides a down-ramp.

Remark: IG provides a sensitivity between local sensitivity and global sensitivity. It plays a role in the preventive control just the same as TS. But the difference is that IG exploits the merit of global sensitivity, which can assess the overall impact of a controllable component on the stability margin, and is free from effect of local stationary points to some extent. Therewith, IG can overcome the issue that local sensitivity is readily to stuck in stationary points. The key problem of the IG idea in the concerned preventive control task is that, since global sensitivity may not be able to know how to make control actions enhance pre-fault transient stability [29], [23], integrated interval of IG cannot be simply set as a global parameter space like global sensitivity dose. It must be carefully designed to allow us to compute control quantities.

III. Integrated Gradients-based Preventive Control

A. Integrated Gradients Based TSCOPF

Following the concept of sensitivity method, TSCOPF is reduced by introducing IG-based linearized constraints, which can be expressed as follows:

1) Objective Function

The controllable variables contain generation of synchronous machines, wind curtailment and loads. The objective is to minimize cost of generation redispatch and load shedding:

$$F_{\text{cost}} = \sum_{i\in\mathbf{S_T}} C_G(P_i) + \sum_{w\in\mathbf{S_W}} C_W(P_w) + \kappa \sum_{l\in\mathbf{S_D}} C_L(P_l) \tag{7}$$

where F_{cost} is the control cost; $\mathbf{S_W}$ and $\mathbf{S_D}$ are respectively the set of wind farms and loads permitted to be shed; κ denotes the permission for load shedding, 1 for load shedding approval, while 0 for prohibition. $C_G(\cdot)$, $C_W(\cdot)$ and $C_L(\cdot)$ are the cost functions of generators, wind farms and load shedding, respectively; P_i, P_w and P_l respectively denote control quantity of the *i*th generator, *w*th wind farm and the *l*th load, and in the context, $\mathbf{P_T} = \{P_i, i \in \mathbf{S}_\text{T}\}, \mathbf{P_W} = \{P_w, w \in \mathbf{S}_\text{w}\}, \mathbf{P_L} = \{P_l, l \in \mathbf{S_D}\}$.

2) Power Flow Constraints

$$\begin{aligned} P_{Gi} &= P_{Li} + \sum_{j=1}^{N_B} |V_i||V_j|(G_{ij}\cos\alpha_{ij} + B_{ij}\sin\alpha_{ij}) \\ Q_{Gi} &= Q_{Li} + \sum_{j=1}^{N_B} |V_i||V_j|(G_{ij}\sin\alpha_{ij} - B_{ij}\cos\alpha_{ij}) \end{aligned}, \forall i, j \in \mathbf{S_B} \tag{8}$$

where P_{Gi} and Q_{Gi} denote active and reactive power generation at bus *i*; P_{Li} and Q_{Li} denote active and reactive load at bus *i*; $|V_i|$ and $|V_j|$ are voltage magnitudes at bus *i* and *j*; α_{ij} is voltage phase angle difference between bus *i* and bus *j*; G_{ij} and B_{ij} are conductance and susceptance located in *i*th row and *j*th column of admittance matrix $\mathbf{Y}$; $\mathbf{S_B}$ is the set of all buses. The nonlinear constraints (8) can be transformed as linear constraints by numerous methods. Since this is not the major concern of this paper, linear form in [30] is directly utilized.

3) Generators Constraints

$$\begin{aligned} P_{Gi}^{\min} \le P_{Gi} \le P_{Gi}^{\max}, \forall i \in \mathbf{S_W} \bigcup \mathbf{S_T} \\ Q_{Gi}^{\min} \le Q_{Gi} \le Q_{Gi}^{\max}, \forall i \in \mathbf{S_W} \bigcup \mathbf{S_T} \end{aligned} \tag{9}$$

where $P_{Gi}^{\max}$ and $P_{Gi}^{\min}$ denote upper and lower action generation of *i*th generator; $Q_{Gi}^{\max}$ and $Q_{Gi}^{\min}$ denote upper and lower reactive generation of *i*th generator.

4) Operational Constraints

$$\begin{aligned} V_i^{\min} \le V_i \le V_i^{\max}, \forall i \in \mathbf{S_B} \\ P_l^{\min} \le P_l \le P_l^{\max}, \forall l \in \mathbf{S_L} \\ P_d^{\min} \le P_d \le P_d^{\max}, \forall d \in \mathbf{S_D} \end{aligned} \tag{10}$$

where $V_i^{\max}$ and $V_i^{\min}$ are upper and lower bounds of voltage at bus *i*, respectively. $P_l^{\max}$ and $P_l^{\min}$ denote upper and lower active power flow of line *l*. $\mathbf{S_L}$ is the set of lines. $P_d^{\min}$ and $P_d^{\max}$ stand for the lower and upper bound of load shedding, respectively.

5) Transient Stability Constraints

Transient stability constrains can be expressed as:

$$\text{TSI} > \text{TSI}_C \tag{11}$$

IG allows to reduce (11) into a simple linearized version:

$$\text{TSI}^0 + \sum_{i\in\mathbf{S_T}\cup\mathbf{S_W}\cup\mathbf{S_D}} \text{IG}_i \times (\Lambda_{C,i} - \Lambda_{C,i}^0) > \text{TSI}_C \tag{12}$$

where TSI^0 and $\Lambda_{\text{C},i}^0$ are respectively the transient stability margin and dispatch at the current operating condition, and IG_i is computed upon the operation. Thus, these parameters are constant. Decision variables are $\mathbf{\Lambda_C}$. Note that, $\Lambda_{\text{C},i} \in \mathbf{\Lambda_C} = [\mathbf{P_T}, \mathbf{P_W}, \mathbf{P_L}]$. TSI_c is a pre-set transient stability threshold.

B. Close-loop and Stepwise Preventive Control Scheme

Model (7)~(12) is a linear TSCOPF problem. To solve it, IPM is introduced, which has been widely testified to be effective to solve constrained optimization problems [14].

During implementation, two issues are noticed. The first is that, despite IG enhances local sensitivity, it is still under a local linearization precondition such that evolution of sensitivity cannot be well followed. The other is how to properly determine integrated interval for IG computation. To address these issues, we propose a stepwise control scheme in a close-loop and adaptive integrated interval tuning framework. The proposed strategy is visualized in Fig. 3 and detailed as follows:

1) Input an operating condition, use an ensemble of multiple DBNs to infer whether the operation is stable or not. If stable, rolling to next operating condition, otherwise, start preventive control;
2) Start from an initial integrated interval $\boldsymbol{\epsilon}^{\mathbf{0}} = [\epsilon_1^0, \epsilon_2^0, \ldots, \epsilon_{N_c}^0]$ (N_c is the total number of controllable resources, e.g., synchronous machines, etc.), which is empirically determined as $\mathcal{B}$% of the rated power of generator or maximum active power of loads that is allowed to be shed ($\mathcal{B}$ is empirically

set to 20~50). Notably, for the *n*th synchronous generator, the initial integrated interval is $[P_n - \epsilon_n^0, P_n + \epsilon_n^0]$, $n \in \mathbf{S_T}$. Yet, for the *w*th wind generator or *l*th load, the integrated intervals are respectively $[P_w - \epsilon_w^0, P_w]$ ($w \in \mathbf{S_W}$) or $[P_l - \epsilon_l^0, P_l]$ ($l \in \mathbf{S_D}$), since wind generation and load are only allowed to be curtailed/shed;

3) Based on the given integrated interval, calculate IGs for all control variables via (6). After that, we get $\mathbf{IG^0}$;
4) Feed the $\mathbf{IG^0}$ to set up transient stability constraint (12), then apply IPM to solve the IG-based TSCOPF to settle initial preventive control actions $\mathbf{\Lambda_C^{P,0}}$ (where we use subscript **C** and superscript **P** to respectively denote the set of control variables and optimized strategy). In this step, IPM taking the symmetric integrated intervals can help the synchronous machines operating at the stationary point escape from zero sensitivity condition. Sensitivities of points near the stationary point can counterbalance each other, resulting in near-zero IG. However, after generation deviates from the stationary point, true sensitivities can therewith go far from 0. Thus, we cannot simply use one step **IG**, otherwise inaccurate preventive control may be induced, appearing as under-control (staying instability) or over-control (rising cost). Next, we introduce an adaptive integrated interval tuning method to fix this problem;
5) According to the initial control actions $\mathbf{\Lambda_C^{P,0}}$, action trends of control variables can be found, so as well as critical generators, which contribute the most efforts in transient stability enhancement [31]. Several top cost-effective units are also involved in the set of critical generators. For efficiency, we only carry out adaptive integrated interval process upon the critical generators. Further, for the generators which increase generation after a step control, we update their integrated intervals to $[P_{T,n} - 0.1\epsilon_n^1, P_{T,n} + \epsilon_n^1]$ such that we can stimulate them to follow the true stability variation trends to improve stability. On the contrary, for those with generation decreased, their integrated intervals change to $[P_{T,n} - \epsilon_n^1, P_{T,n} + 0.1\epsilon_n^1]$. Note that $\boldsymbol{\epsilon^1}$ is the integrated interval at the current 1st iteration. We use (13) to adaptively tune integrated intervals;

$$\boldsymbol{\epsilon}^1 = \frac{\left(\boldsymbol{\epsilon}^0 + \left|\mathbf{\Lambda_C^{P,0}} - \mathbf{\Lambda_C^0}\right|\right)}{2} \tag{13}$$

Notably, (13) is in fact a heuristic search method for integrated interval. It allows us to search control direction throughout nearly the global operational region at the beginning, and conduct the follow-up stepwise control with stable, robust, and enhanced sensitivity information.

6) Calculate IGs with the new integrated intervals, then re-execute IPM, update $\mathbf{\Lambda_C^{P,0}}$ to $\mathbf{\Lambda_C^{P,1}}$, and update $\mathbf{IG^0}$ to $\mathbf{IG^1}$;
7) While $\|\boldsymbol{\epsilon}^k - \boldsymbol{\epsilon}^{k-1}\| > \Delta$, $k \geq 2$ (Δ is preset threshold), or within iteration limit, or system has not turned into stability, perform the loops until the above stop criterion is hit:

$$\boldsymbol{\epsilon}^{k+1} = \frac{\left(\boldsymbol{\epsilon}^k + \left|\mathbf{\Lambda_C^{P,k}} - \mathbf{\Lambda_C^0}\right|\right)}{2} \tag{14}$$

and re-execute IPM, update $\mathbf{\Lambda_C^{P,k}}$ to $\mathbf{\Lambda_C^{P,k+1}}$, and update $\mathbf{IG}^k$ to $\mathbf{IG}^{k+1}$. If IPM does not converge, involve load shedding;
8) Apply the in-use DBN ensemble to assess transient stability. If it is inferred that $\mathrm{TSI}_P < \mathrm{TSI}_\Delta$, the current control actions are under-control [32], then increase conservativeness index TSI_C and re-execute steps 2)-7).

After the proposed stepwise control loop has terminated, perform traditional TSA tool (e.g., TDS or direct method) to verify the efficacy of the control activities.

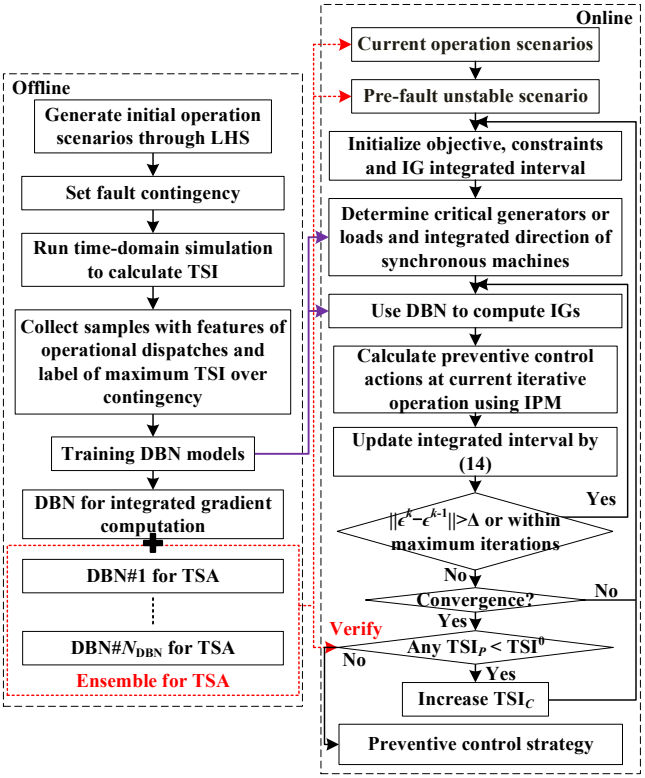


Fig. 3. Proposed close-loop and stepwise integrated gradients-based transient stability preventive control strategy.

IV. Numerical Results

A. Test System and Data Sample Preparation

The proposed method is tested on the modified IEEE 39-bus system. Two wind farms with the rated power of 500MW are connected at bus 17 and 21, respectively. The total installed capacity of wind farms accounts for 16.3% of the total installed capacity (6135MW). To perform transient stability simulation, synchronous machines, governors, power system stabilizers, and wind generators are modelled as 6-order dynamic model, type-I governor, type-I stabilizer and doubly fed induction generator presented in [34], respectively. Contingencies are pre-selected as three-phase (3Ø) short-circuit faults of all transmission lines, in which the fault start time is 0.1s, the fault clearing time is 0.6s, and the simulation duration is 5s. These contingencies will be fed to TDS such that transient stability margin can be computed to label sampled observations, which are generated follow the following perturbation rules. The perturbation envelopes for critical operational variable are set as follows: 80% ~ 120% for active power of synchronous machines $\mathbf{P_T}$ and loads $\mathbf{P_L}$, 1.0 ~ 1.05 p.u. for terminal voltage magnitude of synchronous machines $\mathbf{V_T}$ and wind farms $\mathbf{V_W}$, and 0 p.u. to rated power for active generation of wind farms $\mathbf{P_W}$. Besides, constant power factors are supposed to control reactive loads $\mathbf{Q_L}$. Follow the above sampling rule, 20,000 of data samples are produced, in which 16,000 samples are divided into the training set, 2,000 samples form the validation set, and the rest belongs to the test

set. Note that each sample takes operational variables as inputs and transient stability margin as outputs, so that inputs and outputs are respectively with 62-dimension and 1 dimension.

B. Training DBNs for transient stability assessment and stability margin estimation

We decide a DBN structure with SeLU the activation functions towards our tasks, and training settings with 100 of batch size, 2000 of epochs, and 0.1 of initial learning rate are selected. The loss function is mean square error. In addition, the learning rate decay is used to ensure the training speed and accuracy, the L_2 regularization is used to prevent overfitting, and Adam is chosen as DBN training optimizer [25].

The performance of the trained DBN can be evaluated by our proposed transient stability prediction accuracy A_P and mean absolute error (MAE), shown as (15):

$$A_P = \frac{1}{N_T}\sum\nolimits_{i=1}^{N_T}\left(\mathrm{TSI}_{Pi}\times\mathrm{TSI}_{Ti}>0\right)$$
$$\mathrm{MAE} = \frac{1}{N_T}\sum\nolimits_{i=1}^{N_T}\left|\mathrm{TSI}_{Pi}-\mathrm{TSI}_{Ti}\right| \tag{15}$$

where N_T is the number of the test set samples; TSI_{Pi} and TSI_{Ti} are ith predicted value and true value of TSI respectively. Notably, A_P and MAE are respectively introduced to assess the performance of the DBN on transient stability inference and that on stability margin estimation.

To determine the optimized structure of DBN, we compared the DBN with several learning algorithms, including 3-layer ANN with a hidden structure of 62-50-1, 4-layer stacked auto-encoder (SAE) with a hidden structure of 62-50-20-1, 4-layer DBN (denoted as DBN[4]) with a hidden structure of 62-50-20-1, 5-layer DBN (denoted as DBN[5]) with a hidden structure of 62-50-20-10-1, 6-layer DBN (denoted as DBN[6]) with a hidden structure of 62-50-20-10-5-1, 7-layer DBN (denoted as DBN[7]) with a hidden structure of 62-50-20-10-5-2-1, and 34-layer deep residual network (ResNet) [33]. The comparative outcomes are shown in Tab. I.

TABLE I
COMPARISON OF ACCURACY OF DIFFERENT MODELS

Model	A_P	MAE	Training time
ANN	96.65%	13.91	576s
SAE	96.20%	9.08	698s
DBN[4]	**97.60%**	7.83	693s
DBN[5]	97.25%	7.46	762s
DBN[6]	97.15%	**6.15**	879s
DBN[7]	96.95%	7.13	892s
ResNet	97.05%	24.01	>1h

From Tab. I, we can see that DBN outperforms other learning models. Specially, DBN[4] and DBN[6] result in the highest A_P and the best MAE, respectively. Therefore, multiple DBN[4] is utilized to produce an ensemble to classify transient stability, whilst DBN[6] is applied for integrated gradients.

A sensitivity test is carried out to testify DBNs in DBN[4] ensemble, the results are shown in Tab. II. Notably, for conservative considerations, the accuracy index in Tab. II is the sum of A_P and misclassifications from stability to instability, as such misclassifications will not induce any damaging consequences, but benefit to increase conservativeness.

TABLE II
SENSITIVITY TEST OF THE NUMBER OF DBN[4] IN ENSEMBLE

Model No.	1	2	3	4	5
Accuracy	99.85%	99.50%	100.0%	100.0%	100.0%

From Tab. II, accuracy improves up to 100.0% till model no. 3. In light of low complexity demand, we finally decide to train 3 DBN[4] towards a stability inference ensemble.

C. Integrated gradients analysis

TABLE III
SENSITIVITIES OF STABILITY MARGIN VERSUS POWER GENERATIONS OF GENERATORS AND WIND FARMS IN THE 39-BUS SYSTEM*

Gen. No.	1	2*	**3**	4	**5**	6
Local Sens.	-0.74	-	**0.00**	0.14	**-0.02**	-8.72
DBN[6]-G	-0.09	-	**0.00**	0.09	**0.17**	-10.7
DBN[6]-IG	-0.21	-	**0.34**	0.12	**0.13**	-73.9
Gen. No.	7	8	9	10	17W*	21W*
Local Sens.	-7.90	0.43	-0.03	0.46	-	-
DBN[6]-G	-9.59	-0.10	0.08	-0.31	-0.51	-5.54
DBN[6]-IG	-61.8	0.28	-0.77	0.91	0.57	-53.6

*#2 generator is slack bus. 17W and 21W denote wind farms integrated at bus 17 and bus 21, respectively.

Upon the above analysis, DBN[6] is used to compute integrated gradients according to (6). For a specified unstable operating condition, the comparisons between traditional local sensitivity, local gradients and IGs are given in Tab. III. Note that, since TS method computes $\partial|\delta^{\max}|/\partial P_g, \forall g \in \mathbf{S_G}$, which mismatches our proposed sensitivity of TSI against controllable variables. Thus, to demonstrate availability of the DBN-based sensitivity, numerical sensitivity is conducted in Tab. III (as well as the later Tab. XII). The numerical sensitivity belongs to one-point sensitivity family, and it is computed by $\Delta\mathrm{TSI}/\Delta\boldsymbol{\Lambda_C}$ (where the definitions of $\boldsymbol{\Lambda_C}$ can be found in (4), and the Δ notation denotes small numerical perturbation).

It can be seen from Tab. III that, for generators with small sensitivity, sign mismatches exist, e.g., generators 5. This is because traditional sensitivity can only know local tendency that generation affects transient stability, but IGs provide more global tendencies, and know how generation impacts stability in the vicinity of the current operation. Back to our observations, we take typical generators 3 and 5 as examples and do not consider control cost here:

1) As for generator 5, IG givens the opposite control advise w.r.t. traditional local sensitivity, because there is significant postive sensitivity (say S for intuition) at the operating condition inside this vicinity. Image that there is an instable initial operation as shown in Fig. 4, we execute control loop with traditional local sensitivity (red dotted line), then generator 5 start to increase its generation to improve stability. However, after this action, operation hit the significant negative sensitivity S, then generator 5 will decrease generation. This action may lead the generator back to the initial state, and under the above bad loop, control may stock at local stationary point. Thanks to IG and our devised iterative strategy, control can escape from such local stationary points instantiated in Fig. 4.

2) As for generator 3, Tab. III shows that IG provides a sensitivity of 0.34, while traditional local sensitivity suggests stopping preventive action. In this case, traditional method will delay preventive actions for the generator 3, which in fact does

affect transient stability in global operational domain. Fig. 5 gives a visualization on this circumstance. It is readily to find that initial operation actually performs a positive impact to transient stability in a global perspective. However, traditional local sensitivity cannot aware such tendency. Consequently, outcomes in Tab. III and the above analysis demonstrate that our method behaves abilities of escaping from local stationary point, which have ever been found in any traditional TS methods.

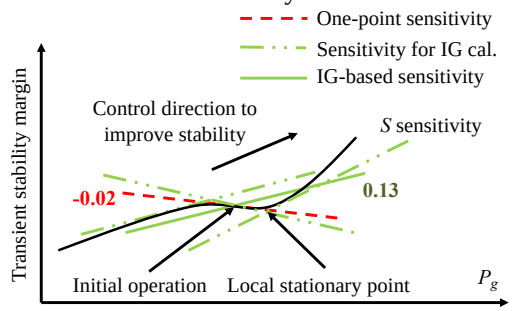


Fig. 4. Visualization of control sensitivities regarding generator 5.

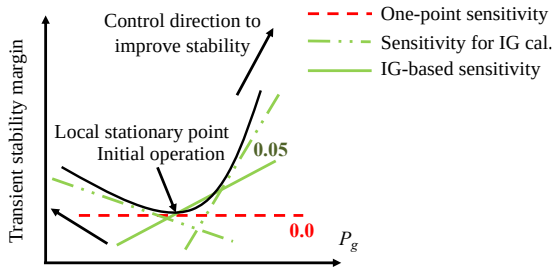


Fig. 5. Visualization of control sensitivities regarding generator 3.

In addition, traditional TS methods only provide control sensitivity of stability against synchronous machines, such that sensitivities for other non-rotating variables (e.g., wind farms, etc.) cannot be derived. This drawback can be solved by integrated gradients, as DBN performs sensitivity calculations for all input variables, including wind generation, load shedding, etc.

To further prove the merit of our method that enables aware of true stability tendency, comparative study of local gradient and integrated gradients is given in Tab. IV, where a generator gradually decreases the power generation by 100 MW and the step is 10 MW.

TABLE IV
COMPARISON OF LOCAL GRADIENT AND INTEGRATED GRADIENT

m	**1**	2	3	4	5
Gradient	**-0.003**	-0.023	-0.208	-2.102	-67.77
m	**6**	**7**	**8**	9	10
Gradient	**-383.1**	**-500.7**	**-390.9**	-50.63	-7.990

Integrated gradient = **-140.4** (according to (6))

Tab. IV depicts how local sensitivity begets imprecise control actions. Since traditional method only takes the local trajectory sensitivity to linearize transient stability constraints, the control sensitivity will remain at -0.003 throughout the control process from the operating condition corresponding to m = 1. In this sense, a 100MW of generator down-ramp will only facilitate a 0.3 enhancement on stability margin. However, this is not true. As you can see from Tab. IV, when power generation reduce by 70MW, true sensitivity turns to -500.7, which significantly diverges from the initial sensitivity of -0.003. With the true sensitivity deploys, only 6e-4MW reduction on power generation is necessary to improve stability margin by 0.3. IG is exactly for fixing this issue. According to (6), IG makes an average with sensitivities on the 10 neighborhoods near the operating condition setpoint, resulting in a -140.4 integrated gradient, which better fits the true stability evolution.

Besides the aforementioned merits w.r.t. control accuracy and ability of escaping from local minimum, DBN-based IG also feature significant efficiency advantage. Detailly, for a specified set of operations, local sensitivity computational mean runtimes via TS and DBN are respectively around 29s and 0.15s. Notably, we have all transmission lines in contingency. For calculation of integrated gradients, 20 times of sensitivity calculations are needed, so you can see traditional methods cannot handle this, but DBN reaches a low runtime around 3s. Also, we can decrease *m* for a faster control demand, and gradient derivation runtime of trained DBN is extremely insensitive to power system scale, which is not the case for TS methods.

Taken together, these outcomes suggest that deep learning-based integrated gradients significantly compromise accuracy and efficiency for practical sensitivity-based transient stability-oriented preventive control. And possible trap in local stationary point by traditional TS methods has been successfully got over via DBN-based IG.

D. Integrated Gradients based Preventive Control

Three different scenarios are designed to testify our methods:

(A) traditional preventive control with only synchronous machine dispatch considered.

(B) traditional preventive control including synchronous machine re-dispatch and wind curtailment.

(C) preventive control considering synchronous machine re-dispatch, wind curtailment and load shedding.

We first test the proposed method in scenario (A). Results are shown in Tab. V.

TABLE V
RESULTS OF THE CONTROL SCENARIO (A)

Gen.	Before Control /MW	After Control /MW	Output change/ MW	Integrated range	Unit control cost /$	Gen. cost /$	Total cost /$
1	**250.4**	**329.8**	**79.4**	**79.5**	**2**	**158.8**	682
2	488.6	488.6	0	0	5	0	
3	**521.5**	**609.7**	**88.2**	**88.8**	**4.5**	**396.9**	
4	585.9	585.9	0	0	9	0	
5	420.6	420.6	0	0	9	0	
6	606.0	606.0	0	0	6	0	
7	**488.6**	**361.9**	**-126.7**	**-126.7**	**1**	**126.7**	
8	439.4	439.4	0	0	5	0	
9	811.9	811.9	0	0	6	0	
10	782.0	782.0	0	0	6	0	

Tab. V shows that preventive actions involved generators 1, 3 and 7, which are just the cost-optimal units among the test system. This supports that the proposed method indeed follows the economic policy to make decisions. Additionally, you can see that our method meets the stop criterion as the controls are almost consistent with the current iterative integrated intervals. This demonstrates our designed iterative control strategy is effective. To further analyze detailed control path, the iteration

process of several generators is given in Tab. VI.

TABLE VI
ITERATION PROCESS OF INTEGRATED RANGES OF CRITICAL GENERATORS

Iteration No.	Integrated range		Power dispatch		TSI by DBN	Cost /$
	Generator No.		Generator No.			
	1	7	1	7		
0	100	100	-	-	**-93.5**	-
1	89.7	-126.0	79.4	-152.0	56.1	**1290**
2	84.5	-126.5	79.4	-127.0	36.7	684
3	81.9	-126.7	79.4	-126.8	36.6	683
4	80.7	-126.7	79.4	-126.7	36.6	683
5	80.0	-126.7	79.4	-126.7	36.5	682
6	79.7	-126.7	79.4	-126.7	36.5	682
7	79.5	-126.7	79.4	-126.7	**36.5**	**682**

Tab. VI givens a control path started with an unstable operation, which features a -93.5 of TSI. It can be observed that the TSI explodes to 56.1 at iteration 0, then progressively decrease to the set TSI_C with cost reduction following. This illustrates that the initial IG steers optimizer to search over global space, and through our adaptive integrated interval tuning, optimizer gradually approaches optimum. This supports the argument that our method fixes the issue of trapping in local stationary point by traditional TS methods. At last, TDS is used to verify the controlled operation. Interestingly, we found that the true TSI is 37.1, which is larger than TSI estimate 36.5. Let's refer back to Table II., You can find the cause of this circumstance is that conservative DBNs are forwarded to drive preventive control. These DBNs are expected to result in the condition that true TSI is better than its estimate. The case shown in Tab. VI supports the above conclusion.

Next, our method is tested in scenario (B), of which the results are shown in Tab. VII.

TABLE VII
RESULTS OF THE CONTROL SCENARIO (B) (INCLUDING WIND CURTAILMENT)

Gen.	Before control /MW	Without wind gen.		With wind gen.	
		After control /MW	Output change /MW	After control /MW	Dispatch power /MW
1	**240.4**	240.4	0	**329.8**	**89.4**
2	521.9	521.9	0	521.9	0
3	**593.1**	593.1	0	**745.1**	**152.0**
4	454.7	454.7	0	454.7	0
5	371.8	371.8	0	371.8	0
6	647.7	647.7	0	647.7	0
7	**588.0**	**441.9**	**-146.1**	588.0	0
8	**555.9**	**706.0**	**150.1**	555.9	0
9	706.3	706.3	0	706.3	0
10	**984.2**	**1019.8**	**35.6**	984.2	0
11	41.9	41.9	0	41.9	0
12	**477.7**	477.7	0	**272.9**	**-204.8**
TSI by DBN		**14.3**		**32.5**	
Cost/$		**1110**		**1067**	

TABLE VIII
RESULTS OF THE CONTROL SCENARIO (C) (INCLUDING LOAD SHEDDING)

Control variable	Before control /MW	After control /MW	Dispatch power/MW	TSI	
				Before control	After control
Gen. 9	888.1	817.8	-70.3	-95.7	46.1
Load 9	770.2	659.7	-110.5		
Cost/$			1747.8		

Tab. VII indicate that the synchronous machine participation for preventive control changed (from generator 7, 8, 10 to generator 1, 3) with wind curtailment involved. This necessities consideration of control sensitivity w.r.t. wind farms, which has not been derived by traditional methods.

Some extreme operations with heavy loads retain limited spinning reserve. The lack of controllable space causes non-existence of preventive control strategy. Towards instability risk-averse, load shedding action has to be dragged in. We then test our method in this scenario (C). Results are shown as Tab. VIII.

Tab. VIII gives the generator and load participating in preventive control, i.e., generator 9 and load 9. In this case, only generator 9 down-ramp reserve for stability promotion. However, after a -70.3 of down regulation, slack generator runs up to its maximum technical output. But the system is still unstable, and there is no resource for further preventive control. Load shedding on load 9 is then performed. Notably, a practical load shedding usually depends on importance rank of loads. Since this is not the focus of this paper, we will not involve that.

E. Comparative study

The traditional TS method deploys local linear sensitivity for nonlinear stability control task, which may cause severe under-control problem. We compare the traditional TS method (TS) [17], our method in scenarios (B) (M(B)) and our method in scenario (C) (M(C)) in several trials. Notably, since the control outcomes of TS method and the numerical sensitivity method are nearly identical, we only report the control effect of TS, as the method is more precise with mathematical theory supported and befits being a preventive control baseline. 231 unstable cases are tested. Results are shown in Tab. IX.

One of the common successful control cases provide that the control costs by TS and M(B)&M(C) are respectively $135 and $52. This outcome and Tab. IX indicate that our method evidently outperforms conventional TS method w.r.t. control success rate, re-dispatch cost and decision efficiency. The first one may be explained by the fact that our method enables escape from local stationary point trap. As for control success rate, TS method cannot follow true nonlinear evolution of stability such that it can be easy to non-convergence. While the proposed IG-based sensitivity and the adaptive interval tuning successfully transfer nonlinearity into iterative control process, such that nonlinearity of stability is maintained, therewith a better economical preventive control is naturally induced. The last improvement (i.e., efficiency) comes from the fast backwards process of deep learning. Therefore, Tab. IX confirms our expectation and justifies the superiority of our method.

TABLE IX
COMPARISON BETWEEN PROPOSED METHOD AND TS[1]

Evaluation index	TS [17]	M(B)	M(C)
Number of successful controls	169	223	228
Number of failed controls	62	8	3
Control success rate	73.2%	96.5%	98.7%
Average cost of successful controls	$1132	$1033	$1180
Average TSI of successful controls	70	49	49
Average running time[2]	261s	28s	31s

1. It is important to note that our experiments are carried out with that all transmission lines are included in contingency set. This is different with common research which only consider one or few contingencies.
2. In this experiment, full ac power flow equations are performed as constraints (8). This enables us to defend the fact that our method features superior efficiency, even when it is used in a challenging nonlinear control problem. In the later numerical studies, we will show that, our approach can achieve significantly quicker decision-making after the convex relaxation trick is used to linearize ac power flow.

F. Verification on a larger system

To further demonstrate the scalability to the larger-scale system, the proposed method is tested in the modified ACTIVSg 200-bus system, which contains 31 synchronous generators, 6 wind farms with type-3 DFIGs, and 160 loads with a total demand of 2229MW and 653MVar [35]. For dynamic models (e.g., generators, governors, stabilizers, etc.), we refer to [35], [36]. We consider credible contingencies as the 3Ø short-circuit faults on the top 10 lines with the longest electrical distance, and set the fault start time, the fault clearing time and the simulation duration as 0.1s, 0.39s, and 10s, respectively. To produce samples, some perturbations are imposed under typical operating condition, i.e., 55% ~ 105% on active power of synchronous machines $\mathbf{P_T}$ and loads $\mathbf{P_L}$, and 0 to rated power on active generation of wind farms $\mathbf{P_W}$. 30,000 of data samples are thus gathered, in which 22,000 samples are randomly picked as training set, 4,000 samples form the validation set, and the rest samples build the test set. Then, a 5-layer DBN (denoted as DBN^5) is developed and trained upon the training set. The structure of the DBN is 360-80-20-10-1. Comparative studies regarding out-of-sample TSA accuracy are provided to reveal the best structure of DL in our task, where DBN^5, DBN^4, DBN^6, DBN^7, ANN and ResNet participated. The outcomes are shown in Table X.

TABLE X
COMPARISON OF ACCURACY OF DIFFERENT MODELS OF 200-BUS SYSTEM

Model	A_P	MAE	Training time
ANN	98.78%	3.20	672s
SAE	97.80%	3.94	516s
DBN^4	98.65%	2.54	613s
DBN^5	**99.38%**	**1.63**	695s
DBN^6	99.00%	2.45	749s
ResNet	98.73%	6.92	>1h

From Tab. X, we can see that compared with other models, DBN^5 has the best A_P and MAE, which is 99.38% and 1.63 respectively. Therefore, the trained DBN^5 will be forwarded to further IG-based sensitivity calculation. For the TSA, three DBN^5 models are combined as DBN ensemble. In addition, for a out-of-sample unstable scenario, the local sensitivity and IGs by DBN^5 are taken into comparisons, where the outcomes are shown in Tab. XI.

TABLE XI
SENSITIVITIES OF STABILITY MARGIN VERSUS POWER GENERATIONS OF GENERATORS AND WIND FARMS IN THE 200-BUS SYSTEM*

Gen. No.	2W	4W	13	14	16	17
Local Sens.	0.28	10.9	1.02	1.35	0.27	0.53
DBN^5-IG	0.34	9.08	1.08	1.63	0.34	-2.12
Gen. No.	18	22	24	26	32	38
Local Sens.	2.25	19.7	-16.0	0.03	0.30	0.29
DBN^5-IG	2.83	31.1	-56.8	-0.01	-0.05	1.53

*2W and 4W denote two wind farms.

What is appealing in Tab. XI is that, there exists some mismatches of gradient direction between the true local sensitivity and IG-based sensitivity. We instantiate the discovery with the sensitivity statistics of the generator 17, where the true TS of 0.53 and the IG-based sensitivity of -2.12 point in different directions for the preventive control. In this case, traditional one-point sensitivity steer generator 17 to increase its power generation, and can be lack of foresight, as neighborhoods around the current operating condition contribute to negative control sensitivities. Similar situation happens on the generator 24, where the IG-based sensitivity of -171.1 is much less than the true TS of -19.4. The outcome implies that neighborhoods around the operating condition of the generator 24 feature smaller control sensitivities, and the generator 24 should not follow the indication of one-point sensitivity, that downwards control its power generation too much. To show the sensitivity distribution of the neighborhoods around the initial operating condition of the generator 24, the local sensitivities and gradients are detailed in Tab. XII.

TABLE XII
ONE-POINT SENSITIVITIES, LOCAL GRADIENTS, AND INTEGRATED GRADIENT AT THE NEIGHBORHOODS AROUND THE OPERATION OF THE GENERATOR 24

m	1	2	3	4	5
Gradient	-19.4	-22.9	-26.0	-23.8	-5.2
Local Sens.	-22.2	-22.2	-27.4	-35.2	-6.7
m	6	7	8	9	10
Gradient	-753.2	-97.4	-57.2	-49.2	-48.2
Local Sens.	-302.4	-140.2	-57.2	-53.2	-51.6

Integrated gradient = **-56.8** (according to (6))

Tab. XII shows that the DBN touches well performance in estimating the one-point sensitivity, as you can see that the local gradients of the DBN are in high accordance with the true sensitivities. The other revealing point is that, there exists a rapid decrease in the control sensitivity at the operating condition of $m = 6$. We highly expect that the situation is induced by nonlinearity of the transient stability. And, apparently, deploying one-point sensitivity at the initial state can be immensely inaccurate under such circumstance.

Upon the above well trained DBN, we perform the preventive control, where synchronous machines and wind generators are involved. A typical unstable scenario is selected to show the control effect. The results are shown in Tab. XIII.

TABLE XIII
OUTCOMES OF THE AC POWER FLOW CONSTRAINED PREVENTIVE CONTROL ON A TYPICAL UNSTABLE OPERATING CONDITION

Gen.	Before con. /MW	After con. /MW	Output change /MW	Integrated range	Gen. cost/$	Total cost/$
8	3.7	4.8	1.1	2.0	5.5	
12	4.0	4.9	0.9	1.1	5.4	
13	26.1	29.3	3.2	3.4	3.2	
15	19.1	29.3	10.2	10.4	61.2	
18	17.3	29.3	12.0	12.2	60	
19	0.8	1.2	0.4	0.6	1.8	518
23	17.2	18.9	1.7	1.9	1.7	
24	150.7	124.9	-25.8	-26.7	129	
27	153.2	206.2	53.0	53.1	106	
32	51.9	72.1	20.2	20.7	121.2	
33	58.3	81.1	22.8	22.9	22.8	

In Tab. XIII, only the synchronous machines which are active in the control are contained. It should be noted that although wind curtailment is involved in this scenario, the IGs of all wind farms are positive. The finding implies that, in this case, wind power penetration benefits transient stability. However, since we cannot upwards control wind power generation beyond the available maximum power, the wind generators maintain their operations. Moreover, you can see that the proposed stepwise control scheme converges when integrated intervals match control actions of the available generators, such that IGs can precisely capture the pivotal impacts of generators on stability.

The elapsed time of online preventive control will be detailed in the subsequent section IV. F.

G. Discussion on computational complexity and cost

In this section, we first discuss how algorithm settings affect control efficiency and cost of our method. The experimental outcomes are given in Tab. XIV.

TABLE XIV
DISCUSSION OF COMPUTATION COST FOR OUR ALGORITHM APPLIED TO THE 200-BUS SYSTEM IN VARIOUS SETTINGS

Algorithm settings	Elapsed time/ s	Control cost/ $	Convergence threshold Δ/ MW
Modelling (8) by AC power flow	212	909	1
Modelling (8) by linearized AC power flow [30]	30	**267**	1
Stopping criterion with only bigger convergence threshold Δ	84	994	20
Stopping criterion with only smaller convergence threshold Δ	425	909	0.5
Stopping criterion with only transient stability being met	**24**	1212	-

*The last three experiments are carried out upon preventive control problem involving AC power flow equalities

Under ideal algorithm settings (i.e., linearized power flow and proper convergence threshold Δ), the proposed method is then contrasted with conventional TS method [15], and recent state-of-the-art surrogate-assisted method [10], regarding computational cost. Specifically, computational complexity theory is introduced to prove merit in efficiency of our method versus conventional TS method, and numerical studies are carried out to compare computational cost between our method and surrogate-assisted method.

We note that, traditional TS method and our algorithm are both part of sensitivity family, which characterizes similar stepwise control strategy, thus their time complexities at each control step can be used as evidence for the contrast of computational complexity. According to the analysis in Appendix, the stepwise time complexity of our method is at least $\mathcal{O}(1/\Delta t^4)$ better than that of TS, where Δt is the timestep of TDS.

Since the nonconvexity and nonlinearity of preventative control make it challenging to estimate total computational complexity of the applied surrogate-assisted method, and solution quality can contradict with total computational complexity, we prefer to compare the computational costs of the applied sensitivity methods and surrogate-assisted method by numerical tests. The experimental outcomes are given in Tab. XV.

TABLE XV
COMPARISON AMONG THE APPLIED METHODS IN TERMS OF CONTROL EFFICIENCY AND CONTROL COST (WITHOUT LOAD SHEDDING)

Methods	Ave. decision runtime	Ave. control cost
The proposed	31s	$243
TS [17]	181s	$289
Surrogate modelling [12]	67s	$261

From Tab. XV, on average, our method outperforms other baselines in terms of decision efficiency and control cost. The outcomes show tremendous potential of DL applying in engineering, and our method successfully exploits computational merits of DL to feedback more information to improve control.

H. Discussion on transient stability indicator selection

In general, there are two mainstreams for building transient stability indicators, which are time-domain simulation (TDS), transient energy function (TEF).

On one hand, TEF-based indicators require only few computational efforts, however, they feature some drawbacks [37]. For example, the computation of these indicators demands strong assumptions, and can loss accuracy and reliability when applying in a bulk power system or a topology varying system. Also, one can be hard to determine a proper Lyapunov function that can precisely draw stability boundary. For preventive control, precision and reliability are the crucial concentrations, so we do not employ TEF-based indicators.

On the other hand, the indicator upon TDS is more accurate but are also time-consuming to compute. These indicators include critical clearing time (CCT) [38], TSI, etc. The TSI is a widespread index in power industries, as it is easy to compute and has been justified to be reliable in engineering. In this paper, we have shown that, taking the TSI as the quantitative indicator, the applied method manifests well performance on estimating stability margin and preventive control. You can find the convincing outcomes in the trials of the IEEE 39-bus and the new ACTIVSg 200-bus system. Compared with the TSI, the CCT is more numerically stable as it can produce a sample distribution with lower variance. However, in our preceding research [38], we had found that, the difference between the estimation error of CCT and that of the TSI is negligibly tiny. But given that CCT is much more computationally intensive than TSI, we prefer to use TSI to quantify the stability margin.

V. CONCLUSION

In this paper, an analytic deep learning and iteratively integrated gradients-based paradigm for improving practical TS method is proposed for power system transient stability preventive control. The key is that fast backwards process of well-trained deep learning is introduced to surrogate computationally intensive traditional TS analysis. In detail, the integrated gradients that come from deep learning backwards are forwarded as control sensitivities to linearize transient stability constraints, such that optimizer is enabled to quickly solve the constrained model. Then, a stepwise control strategy is proposed to cooperate integrated gradients and optimizer, which allows us to follow true variation tendency of TSI to generate control actions. In addition to the nature efficiency gain via the proposed method, drawbacks of inaccurate control and local minimum from local linear sensitivity are well improved by the integrated gradients. The numerical outcomes on the modified IEEE 39-bus system and ACTIVSg 200-bus system justify that, compared to traditional TS method, the proposed method can significantly improve under-control and over-control (in other words, reduce control cost), and also unfolds an efficiency

merit. Notably, our method follows the engineering idea of sensitivity and is compatible with existing preventive control tools, so it holds a promising engineering practical prospect.

APPENDIX

A. Computational complexity approximation for local TS and the proposed IG-based sensitivity

1) As for the computational complexity analysis of our method, we first subdivide our method into three key modules, which are stepwise nonlinear programming module, transient stability assessment module, and transient stability linearization module based on integrated gradients. The computational complexity of each module will be then discussed in detail.

i) stepwise computational complexity of our algorithm:

Stepwise linear programming module: since interior point method (IPM) is conducted to solve the linearized TSCOPF, the computational complexity is just the same with the nature complexity of IPM algorithm. Thereby, for the computation complexity of total iterations, this module features $\mathcal{O}(n^{0.5}\log(1/\epsilon))$, where n and ϵ are the dimension of decision variables and convergence threshold of duality gap, respectively. Then, considering that integrated gradient method linearizes transient stability constraints, the model (7)~(12) is a typical linear programming. Thus, for Newton solver at every IPM's iteration, computational complexity of $\mathcal{O}(np^3+p^2n^2+n^2)$ holds, where p is the number of constraints.

TSA module: suppose we have well trained N_{DBN} DBNs for ensemble, of which the ith one owns a structure of $|\mathbf{\Lambda}|$-H_1^i-H_2^i-$\cdots$-$H_{\mathcal{L}-1}^i$-1 (the number of hidden neurons from the input layer to the output layer is indicated by elements from left to right in order, and $|\mathbf{\Lambda}|$ also denotes the dimension of the input features. $\mathcal{L}$ is the quantity of DBN layers). Then, computational complexity for transient stability assessment is $\mathcal{O}\left(\sum_{k=1}^{N_{\mathrm{DBN}}}\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k\right)$, where $H_{\mathcal{L}}^k=1$, and here, $H_0^k=|\mathbf{\Lambda}|$.

Transient stability linearization module: in this paper, integrated gradient (IG) method is conducted to linearize transient stability constraints, so complexity of IG computation should be involved in overall complexity analysis. In general, a DBN can be represented as (R1).

$$\begin{cases} O_1=\mathcal{A}\big(\mathcal{M}(\mathbf{\Lambda})\big)\\ O_2=\mathcal{A}\big(\mathcal{M}(O_1)\big)\\ \quad\vdots\\ O_{\mathcal{L}-1}=\mathcal{A}\big(\mathcal{M}(O_{\mathcal{L}-2})\big)\\ \mathcal{R}^{\boldsymbol{\theta}^*}(\mathbf{\Lambda})=\Gamma(O_{\mathcal{L}-1}) \end{cases} \tag{16}$$

where $\mathcal{M}(\cdot),\mathcal{A}(\cdot)$ and $\Gamma(\cdot)$ are affine function, activation function, and output function, respectively. $\boldsymbol{\theta}^*$ denotes well trained structural parameters of DBN.

Upon (16), control sensitivity of transient stability margin against controllable variables ($\mathbf{\Lambda_C}\subset\mathbf{\Lambda}$) can be derived as:

$$\frac{\partial\mathcal{R}^{\boldsymbol{\theta}^*}(\mathbf{\Lambda})}{\partial\mathbf{\Lambda_C}}=\frac{\partial\Gamma(O_{\mathcal{L}-1})}{\partial O_{\mathcal{L}-1}}\frac{\partial O_{\mathcal{L}-1}}{\partial\mathbf{\Lambda_C}} \tag{17}$$

$$\frac{\partial O_{\mathcal{L}-1}}{\partial\mathbf{\Lambda_C}}=\frac{\partial\mathcal{A}\big(\mathcal{M}(O_{\mathcal{L}-2})\big)}{\partial\mathcal{M}(O_{\mathcal{L}-2})}\circ\frac{\partial\mathcal{M}(O_{\mathcal{L}-2})}{\partial O_{\mathcal{L}-2}}\frac{\partial O_{\mathcal{L}-2}}{\partial\mathbf{\Lambda_C}} \tag{18}$$

$$\vdots$$

$$\frac{\partial O_2}{\partial\mathbf{\Lambda_C}}=\frac{\partial\mathcal{A}\big(\mathcal{M}(O_1)\big)}{\partial\mathcal{M}(O_1)}\circ\frac{\partial\mathcal{M}(O_1)}{\partial O_1}\frac{\partial O_1}{\partial\mathbf{\Lambda_C}} \tag{19}$$

$$\frac{\partial O_1}{\partial\mathbf{\Lambda_C}}=\frac{\partial\mathcal{A}\big(\mathcal{M}(\mathbf{\Lambda})\big)}{\partial\mathcal{M}(\mathbf{\Lambda})}\circ\frac{\partial\mathcal{M}(\mathbf{\Lambda})}{\partial\mathbf{\Lambda}}\frac{\partial\mathbf{\Lambda}}{\partial\mathbf{\Lambda_C}} \tag{20}$$

where $\circ$ denotes Hadamard product operator.

Just execute backward computation from (20) to (17), control sensitivity can be got. Note that, in $\partial\mathbf{\Lambda}/\partial\mathbf{\Lambda_C}$ of (20), only the entries corresponding to $\mathbf{\Lambda_C}$ are 1 and others are 0, since $\mathbf{\Lambda_C}\subset\mathbf{\Lambda}$. These zero elements will not impact computational complexity, so that the dimension of $\partial\mathbf{\Lambda}/\partial\mathbf{\Lambda_C}$ can be deemed as N_c (the number of control variables), and $\partial\mathcal{M}(\mathbf{\Lambda})/\partial\mathbf{\Lambda}\in\mathbb{R}^{H_1^i\times N_c}$ according to the same evidence. In light of this, computational complexity for DBN-based control sensitivity calculation is $\mathcal{O}\left(\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k\right)$, where $H_0^k=N_c$ here. Combined with IG computing formula (6) in the manuscript, transient stability linearization module holds computational complexity of $\mathcal{O}\left(m\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k\right)$ (m is the number of interpolation points over an integrated interval). Usually, $m\ll\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k$, so $\mathcal{O}\left(m\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k\right)\approx\mathcal{O}\left(\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k\right)$. It is discovered that, the complexity is trivial in contrast to the computational complexity of transient stability linearization module, such that we may disregard it while computing overall complexity.

Stepwise computational complexity of our algorithm: the above analysis provides that our method features a computational complexity of (21):

$$\mathcal{O}\big((np^3+p^2n^2+n^2)n^{0.5}\log(1/\epsilon)\big)+\mathcal{O}\left(\sum_{k=1}^{N_{\mathrm{DBN}}}\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k\right) \tag{21}$$

For a power system application, $\mathcal{O}\big((np^3+p^2n^2+n^2)n^{0.5}\log(1/\epsilon)\big)$ is usually much larger than $\mathcal{O}\left(\sum_{k=1}^{N_{\mathrm{DBN}}}\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k\right)$. Based on this fact, we have

$$\mathcal{O}\big((np^3+p^2n^2+n^2)n^{0.5}\log(1/\epsilon)\big)+\mathcal{O}\left(\sum_{k=1}^{N_{\mathrm{DBN}}}\sum_{i=1}^{\mathcal{L}}H_i^k\,H_{i-1}^k\right)<2\mathcal{O}\big((np^3+p^2n^2+n^2)n^{0.5}\log(1/\epsilon)\big)=\mathcal{O}\big((np^3+p^2n^2+n^2)n^{0.5}\log(1/\epsilon)\big),$$

Therefore, the computational complexity of the whole algorithm is nearly the same with that of the stepwise nonlinear programming, as shown in (22).

$$\mathcal{O}\big((np^3+p^2n^2+n^2)n^{0.5}\log(1/\epsilon)\big) \tag{22}$$

We further compare the computational complexity between our method and several related studies, which comprise TS method proposed in [15]. Other TS methods only improve subtle computational complexity, so we will not discuss them).

ii) stepwise computational complexity of traditional TS method:

The major discrepancy between conventional TS method and our technique is control sensitivity computation that is used for linearization of transient stability constraints. In this sense, by simply re-deriving control sensitivity computational complexity, we can attain the computational complexity of TS.

TS is a byproduct of time-domain simulation, it therewith necessitates trajectory simulation up till appearance of an out-of-step generator, of which a precise computation asks for fully dynamic modelling of governors, excitation systems, power system stabilizers, electric motors, synchronous machines, and

so on. Despite the computational complexity of this process is hard to analyze and is undoubtedly higher than that of our algorithm, we attempt to declare it with only simple 2-order synchronous machine models involved.

$$\dot{\delta}_i = \omega_i - \omega_N = f(\boldsymbol{\omega}), \forall i \in \mathbf{S_T} \tag{23a}$$

$$M_i\dot{\omega}_i = \omega_N(-D_i\omega_i + P_{mi} - P_{ei}) = g(\boldsymbol{\omega}, \boldsymbol{\delta}) \tag{23b}$$

(23) is not analyzable in practice, and numerical algorithm is prevailing to solve (23). To attain numerical solution, (23) will be converted into difference equations by implicit integration rules. For instance, a common 4th order Runge-Kutta method renders (24) to algebraize (23).

$$\delta_i^{(t)} = \delta_i^{(t-1)} + \frac{\Delta t}{6}\left(f\left(\boldsymbol{\omega}^{(t-1)}\right) + 2f\left(\begin{array}{c}\boldsymbol{\omega}^{(t-1)} + \frac{\Delta t f\left(\boldsymbol{\omega}^{(t-1)}\right)}{2} + \frac{\Delta t f\left(\frac{\Delta t f\left(\boldsymbol{\omega}^{(t-1)}\right)}{2}\right)}{2} + \\ \Delta t f\left(\frac{\Delta t f\left(\frac{\Delta t f\left(\boldsymbol{\omega}^{(t-1)}\right)}{2}\right)}{2}\right)\end{array}\right)\right) \tag{24}$$

The time complexity of the 4th order Runge-Kutta is $\mathcal{O}(1/\Delta t^4)$. Note that, in every timestamp of Runge-Kutta method implementation, electrical power P_{ei} should be updated via solving algebraic equations, which are augmented version of steady-state power flow equations with fault and internal impedance of generators contained. To solve the algebraic equations, Newton-Raphson method is conducted, of which the computational complexity is approximated as $\mathcal{O}(\log(|\mathbf{S_B}|))$. Then computational complexity of TS can be estimated below.

$$\mathcal{O}\left((np^3 + p^2n^2 + n^2)n^{0.5}\log(1/\epsilon)\right) + \mathcal{O}(1/\Delta t^4) + \mathcal{O}(\log(|\mathbf{S_B}|)) \tag{25}$$

Since $\mathcal{O}(\log(|\mathbf{S_B}|))$ is much less than the first two, we can finally get computational complexity for TS method, shown as (26). Note that Δt is usually pre-set as 1~3 cycles of power systems, i.e., 0.02s~0.06s, thus $\mathcal{O}(1/\Delta t^4)$ is nonnegligible with regard to the first term.

$$\mathcal{O}\left((np^3 + p^2n^2 + n^2)n^{0.5}\log(1/\epsilon)\right) + \mathcal{O}(1/\Delta t^4) \tag{26}$$

As clarified in (22) and (26), our algorithm beats conventional TS methods with regard to computational complexity.

REFERENCES

[1] J. Zhao *et al.*, "Roles of dynamic state estimation in power system modeling, monitoring and operation," *IEEE Trans. Power Syst.*, vol. 36, no. 3, pp. 2462–2472, May 2021,

[2] S. Zadkhast, J. Jatskevich, and E. Vaahedi, "A multi-decomposition approach for accelerated time-domain simulation of transient stability problems," *IEEE Trans. Power Syst.*, vol. 30, no. 5, pp. 2301–2311, Sep. 2015,

[3] H. Yuan and Y. Xu, "Preventive-corrective coordinated transient stability dispatch of power systems with uncertain wind power," *IEEE Trans. Power Syst.*, vol. 35, no. 5, pp. 3616–3626, Sep. 2020,

[4] W. S. Rosenthal, A. M. Tartakovsky, and Z. Huang, "Ensemble kalman filter for dynamic state estimation of power grids stochastically driven by time-correlated mechanical input power," *IEEE Trans. Power Syst.*, vol. 33, no. 4, pp. 3701–3710, Jul. 2018,

[5] C. Mishra, A. Pal, J. S. Thorp, and V. A. Centeno, "Transient stability assessment of prone-to-trip renewable generation rich power systems using lyapunov's direct method," *IEEE Trans. Sustain. Energy*, vol. 10, no. 3, pp. 1523–1533, Jul. 2019,

[6] T. Xiao, J. Wang, Y. Gao, and D. Gan, "Improved sparsity techniques for solving network equations in transient stability simulations," *IEEE Trans. Power Syst.*, vol. 33, no. 5, pp. 4878–4888, Sep. 2018,

[7] J. Lv, M. Pawlak, and U. D. Annakkage, "Prediction of the transient stability boundary based on nonparametric additive modeling," *IEEE Trans. Power Syst.*, vol. 32, no. 6, pp. 4362–4369, Nov. 2017,

[8] F. R. Gomez, A. D. Rajapakse, U. D. Annakkage, and I. T. Fernando, "Support vector machine-based algorithm for post-fault transient stability status prediction using synchronized measurements," *IEEE Trans. Power Syst.*, vol. 26, no. 3, pp. 1474–1483, Aug. 2011,

[9] X. Ye and J. V. Milanovic, "Composite index for comprehensive assessment of power system transient stability," *IEEE Trans. Power Syst.*, pp. 1–1, 2021,

[10] T. Su, Y. Liu, J. Zhao, and J. Liu, "Deep belief network enabled surrogate modeling for fast preventive control of power system transient stability," *IEEE Trans. Ind. Inform.*, vol. 18, no. 1, pp. 315–326, Jan. 2022,

[11] T. Su, Y. Liu, J. Zhao, and J. Liu, "Probabilistic stacked denoising autoencoder for power system transient stability prediction with wind farms," *IEEE Trans. Power Syst.*, vol. 36, no. 4, pp. 3786–3789, Jul. 2021,

[12] A. Gupta, G. Gurrala, and P. S. Sastry, "An online power system stability monitoring system using convolutional neural networks," *IEEE Trans. Power Syst.*, vol. 34, no. 2, pp. 864–872, Mar. 2019,

[13] H. Yuan, Y. Xu, and C. Zhang, "Robustly coordinated generation dispatch and load shedding for power systems against transient instability under uncertain wind power," *IEEE Trans. Power Syst.*, pp. 1–1, 2021,

[14] Quanyuan Jiang and Guangchao Geng, "A reduced-space interior point method for transient stability constrained optimal power flow," *IEEE Trans. Power Syst.*, vol. 25, no. 3, pp. 1232–1240, Aug. 2010,

[15] T. B. Nguyen and M. A. Pai, "Dynamic security-constrained rescheduling of power systems using trajectory sensitivities," *IEEE Trans. Power Syst.*, vol. 18, no. 2, pp. 848–854, May 2003,

[16] G. Qiu *et al.*, "Analytic deep learning-based surrogate model for operational planning with dynamic ttc constraints," *IEEE Trans. Power Syst.*, vol. 36, no. 4, pp. 3507–3519, Jul. 2021,

[17] C. Liu *et al.*, "A systematic approach for dynamic security assessment and the corresponding preventive control scheme based on decision trees," *IEEE Trans. Power Syst.*, vol. 29, no. 2, pp. 717–730, Mar. 2014,

[18] J. Sun, G. Qi, N. Mazur, and Z. Zhu, "Structural scheduling of transient control under energy storage systems by sparse-promoting reinforcement learning," *IEEE Trans. Ind. Inform.*, vol. 18, no. 2, pp. 744–756, Feb. 2022,

[19] I. Genc, R. Diao, V. Vittal, S. Kolluri, and S. Mandal, "Decision tree-based preventive and corrective control applications for dynamic security enhancement in power systems," *IEEE Trans. Power Syst.*, vol. 25, no. 3, pp. 1611–1619, Aug. 2010,

[20] M. C. Passaro, A. P. A. da Silva, and A. C. S. Lima, "Preventive control stability via neural network sensitivity," *IEEE Trans. Power Syst.*, vol. 29, no. 6, pp. 2846–2853, Nov. 2014.

[21] Ginocchi, Mirko, Ferdinanda Ponci, and Antonello Monti, "Sensitivity Analysis and Power Systems: Can We Bridge the Gap? A Review and a Guide to Getting Started," in *Energies*, vol. 14, no. 24, pp. 8274, 2021.

[22] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proceedings of the 34th International Conference on Machine Learning*, Jul. 2017, pp. 3319–3328.

[23] A. Pizano-Martínez, C. R. Fuerte-Esquivel, E. A. Zamora-Cárdenas and J. M. Lozano-García, "Directional Derivative-Based Transient Stability-Constrained Optimal Power Flow," in *IEEE Transactions on Power Systems*, vol. 32, no. 5, pp. 3415-3426, Sept. 2017

[24] K. Ye, J. Zhao, C. Huang, N. Duan, Y. Zhang and T. E. Field, "A Data-Driven Global Sensitivity Analysis Framework for Three-Phase Distribution System With PVs," in *IEEE Transactions on Power Systems*, vol. 36, no. 5, pp. 4809-4819, Sept. 2021.

[25] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, Jul. 2006.

[26] H. Sun *et al.*, "Automatic learning of fine operating rules for online power system security control," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 8, pp. 1708–1719, Aug. 2016.

[27] H. Zhao, A. P. Sinha, G. Bansal, "An extended tuning method for cost-sensitive regression and forecasting," in *Decision support systems*, vol. 51, no. 3, pp. 362-383, 2011

[28] J.-S. Pan, N. Liu, S.-C. Chu, and T. Lai, "An efficient surrogate-assisted hybrid optimization algorithm for expensive optimization problems," *Inf. Sci.*, vol. 561, pp. 304–325, Jun. 2021.

[29] F. Ni, M. Nijhuis, P. H. Nguyen and J. F. G. Cobben, "Variance-Based Global Sensitivity Analysis for Power Systems," in *IEEE Transactions on Power Systems*, vol. 33, no. 2, pp. 1670-1682, March 2018.

[30] S. M. Fatemi, S. Abedi, G. B. Gharehpetian, S. H. Hosseinian and M. Abedi, "Introducing a Novel DC Power Flow Method With Reactive Power Considerations," in *IEEE Transactions on Power Systems*, vol. 30, no. 6, pp. 3012-3023

[31] R. Yan, G. Geng, and Q. Jiang, “Data-driven transient stability boundary generation for online security monitoring,” *IEEE Trans. Power Syst.*, vol. 36, no. 4, pp. 3042–3052, Jul. 2021,

[32] G. Qiu, Y. Liu, J. Liu, et al., “Surrogate-assisted optimal re-dispatch control for risk-aware regulation of dynamic total transfer capability,” in *IET Gener. Transm. Distrib*., vol. 15, pp. 1949–1961, 2021.

[33] K. Chen, K. Chen, Q. Wang, Z. He, J. Hu, and J. He, “Short-term load forecasting with deep residual networks,” *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 3943–3952, Jul. 2019.

[34] Milano, F (2010, May), "Power system analysis toolbox 2.1.6.," University College Dublin, Ireland. [Online]. Available: http://faraday1.ucd.ie/psat.html#download.

[35] “Electric Grid Test Case Repository - Synthetic Electric Grid Cases,” [Online]. Available: https://electricgrids.engr.tamu.edu/.

[36] T. Xu, A. B. Birchfield and T. J. Overbye, "Modeling, Tuning, and Validating System Dynamics in Synthetic Electric Grids," in *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6501-6509, Nov. 2018.

[37] Sobbouhi, Ali Reza, and Abolfazl Vahedi., "Transient stability prediction of power system; a review on methods, classification and considerations," in *Electric Power Systems Research*, vol. 190, pp. 106853, 2021.

[38] T. Liu, et al., "Multi-indicator inference scheme for fuzzy assessment of power system transient stability," in *CSEE Journal of Power and Energy Systems*, vol. 2, no. 3, pp. 1-9, Sept. 2016